

Background

Humans continually adapt their beliefs to changing environments by updating predictions when outcomes violate expectations. These mismatches generate **prediction errors (PEs)**, but the size of the prediction error alone does not determine how much people should update their beliefs. Instead, humans must infer the underlying **temporal structure** of the environment and adjust their **learning rate (LR)** accordingly.

In **oddball environments (OB)**, unexpected outcomes are likely random fluctuations, and thus large prediction errors should produce relatively small updates. In contrast, in **change point environments (CP)**, unexpected outcomes may signal a shift in the environment, requiring larger learning rates and rapid belief updating.

Previous work shows that humans adapt learning rates to different temporal structures. However, sequential experience biases subsequent learning [1], with prior exposure to change point environments producing persistent over-updating in later oddball environments [2].

Objective:

Does resetting prior expectations reduce sequential bias? What are the cognitive mechanisms underlying this effect?

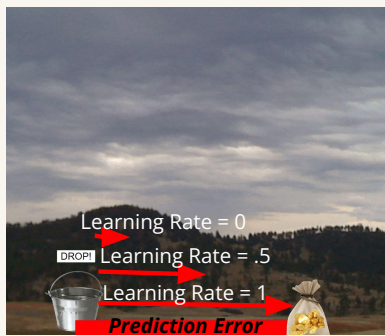
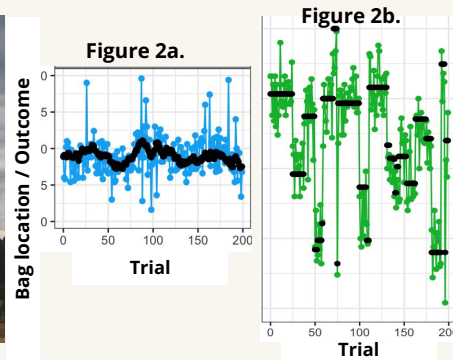


Figure 1. Helicopter Prediction Task



Changepoint (green):

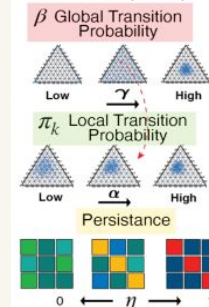
Outcomes shift to a new location and remain there; large errors signal lasting change.

Oddball (blue): Outcomes remain centered around a stable location with occasional outliers; large errors are temporary and should be downweighted.

Computational Model: Behavior was simulated using a hierarchical Bayesian particle-filter model. The model contained 2,400 particles spanning every combination of the parameter grid ($4a \times 6\eta \times 5\gamma \times 5$ (drift-variance) $\times 4$ (observation-noise) values). Each particle represented one candidate parameter combination and was weighted according to how well it predicted the observed outcomes.

Figure 3.

Roles of γ , α , and η in the model's transition structure, adapted from Ramzi et al. (2026).

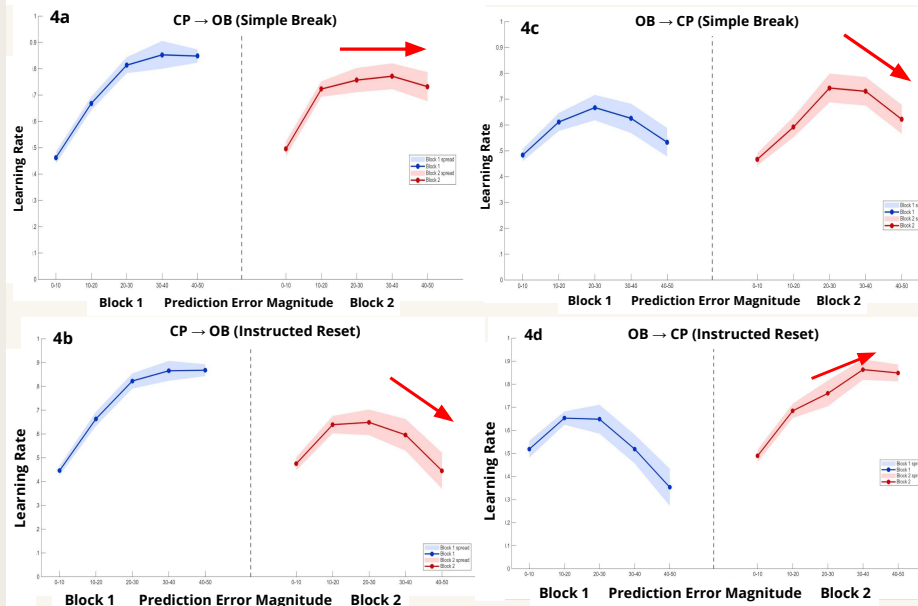


Block-transition manipulations:

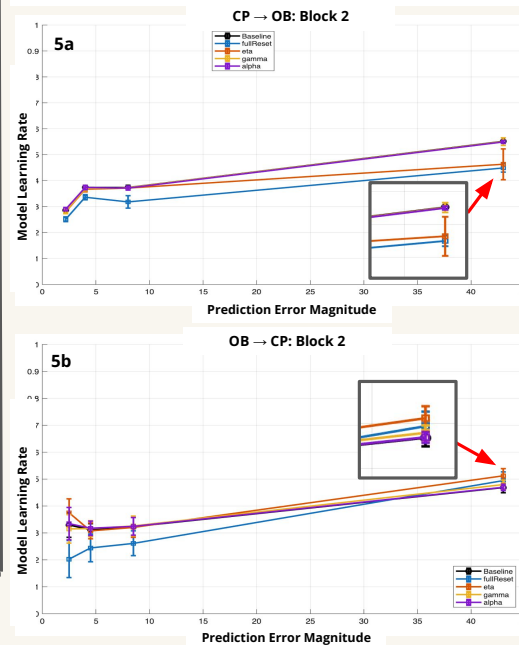
- **Baseline:** No manipulation
- **Full reset:** Particle log-likelihoods were reset, giving all 2,400 parameter combinations equal weight while retaining each particle's learned internal state.
- **Selective reset:** The probability distribution over α , η , or γ was flattened, removing evidence favoring particular values of the targeted parameter

Results

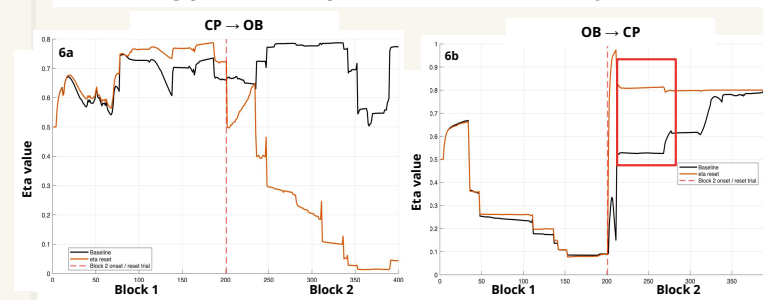
Humans carry over learned environmental structure, reset cues reduce this bias



Computational Modeling: Parameter Resets did not significantly alter learning rates



Resetting persistence parameter reduces carryover effects



Conclusion

- Prior learning experience biased subsequent learning when the learned temporal structures are distinct. The instructed reset manipulation successfully reduced this carryover effect in human participants.
 - Resetting η showed promising early adaptation to the new environment and is a potential candidate explanation of the effect of instructed reset.
- Future Directions
- We reset only one of five hyperparameters at a time. We can test joint resets of all 10 hyperparameter pairs and 10 triplets to identify combinations that improve adaptation.